Diffusion models

The forward and backward process, training, and conditioning

Carlos Cotrini

ETH Zurich

21 October 2026

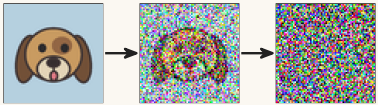
Diffusion models in brief

A diffusion model generates new data, such as images, by learning to undo a process that gradually adds noise.

Diffusion models in brief

A diffusion model generates new data, such as images, by learning to undo a process that gradually adds noise.

Training

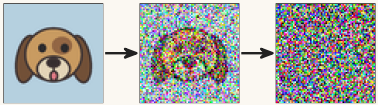


Take real images, add noise in many small steps, and learn to predict the noise that was added.

Diffusion models in brief

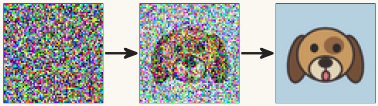
A diffusion model generates new data, such as images, by learning to undo a process that gradually adds noise.

Training



Take real images, add noise in many small steps, and learn to predict the noise that was added.

Generation

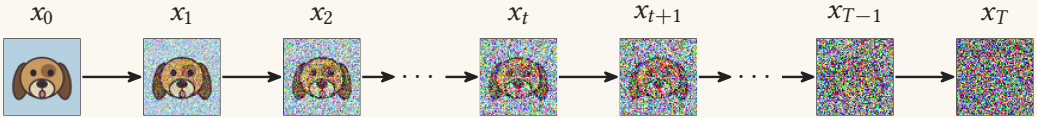


Start from pure noise and remove the predicted noise, step by step, until a clean image appears.

Forward & backward process

Each forward step fades the image and adds noise; generation runs the chain backwards.

fade & noise

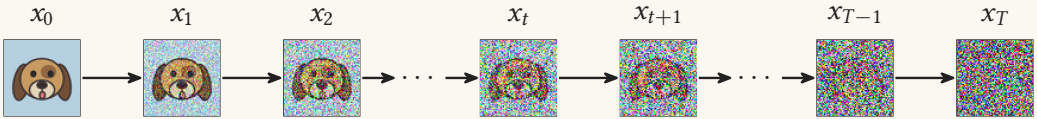


Forward & backward process

Each forward step fades the image and adds noise; generation runs the chain backwards.

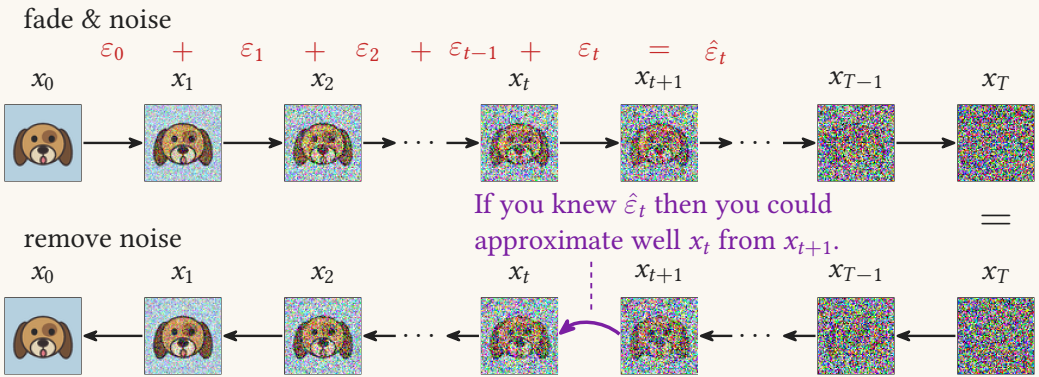
fade & noise

$\varepsilon_0 \quad + \quad \varepsilon_1 \quad + \quad \varepsilon_2 \quad + \quad \varepsilon_{t-1} \quad + \quad \varepsilon_t \quad = \quad \hat{\varepsilon}_t$



Forward & backward process

Each forward step fades the image and adds noise; generation runs the chain backwards.



Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

1. Take an image of the dataset.

x_0



Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

1. Take an image of the dataset.
2. Pick $t \leq T$ at random.

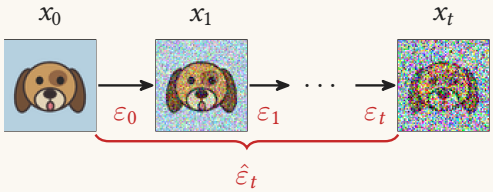
x_0



Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

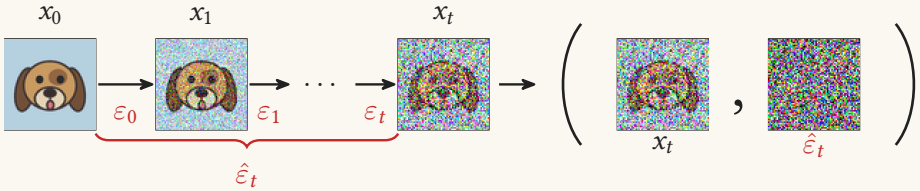
1. Take an image of the dataset.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

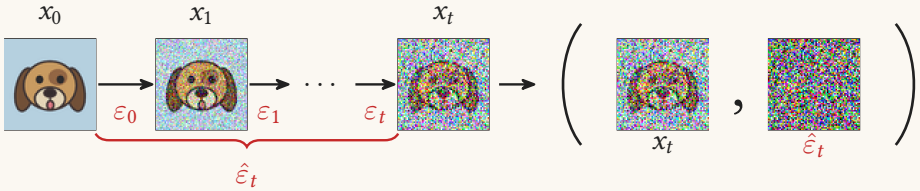
1. Take an image of the dataset.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Training of a diffusion model

The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

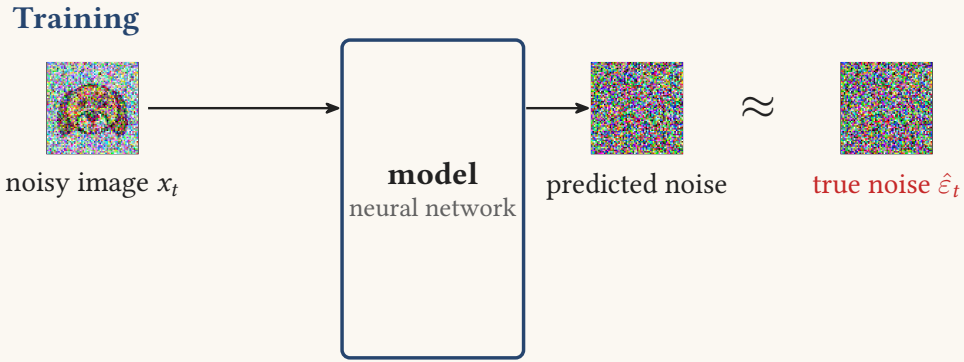
1. Take an image of the dataset.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Repeat until you collect $\{(x_t^{(1)}, \hat{\varepsilon}_t^{(1)}), \ldots, (x_t^{(n)}, \hat{\varepsilon}_t^{(n)})\}$

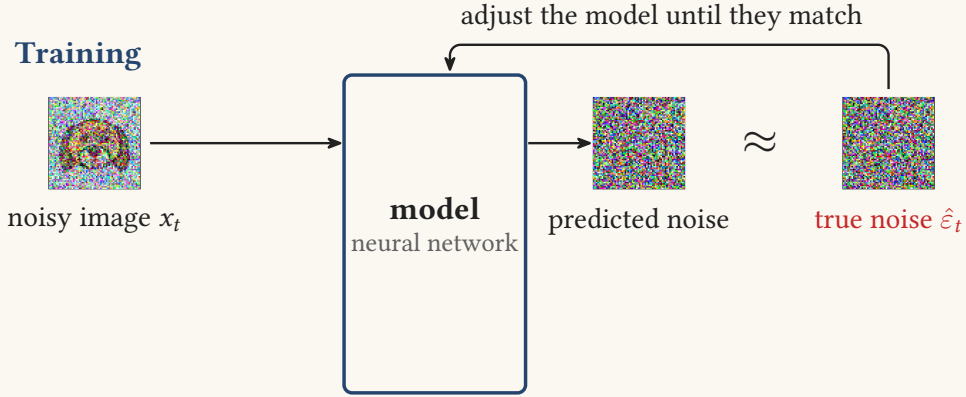
What the model learns

One neural network, two uses: trained to predict the noise, then used to remove it.



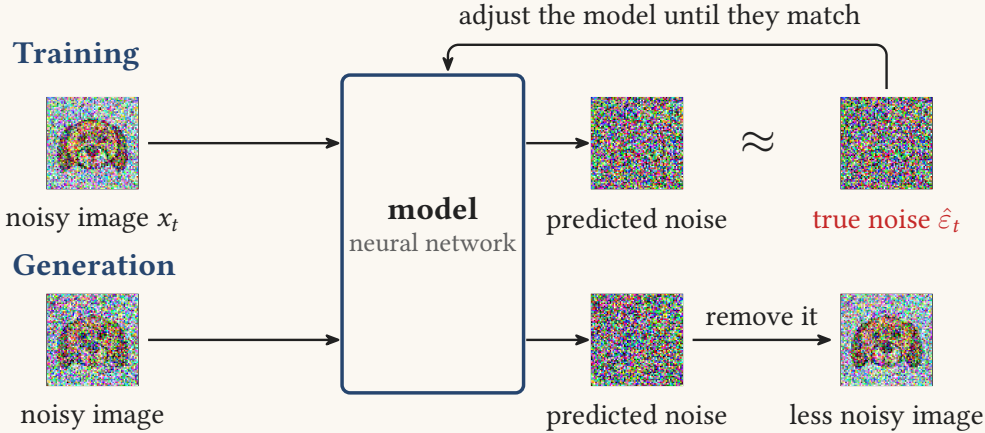
What the model learns

One neural network, two uses: trained to predict the noise, then used to remove it.



What the model learns

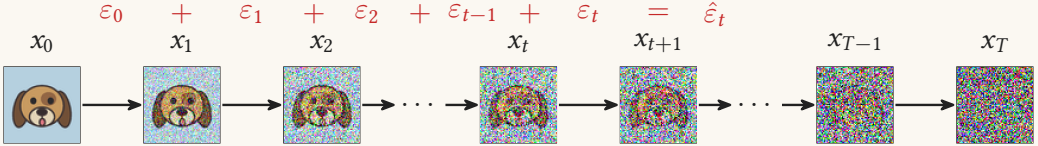
One neural network, two uses: trained to predict the noise, then used to remove it.



Conditional diffusion models

The prompt "cat" is given to every backward step, so the image it produces is a cat.

fade & noise

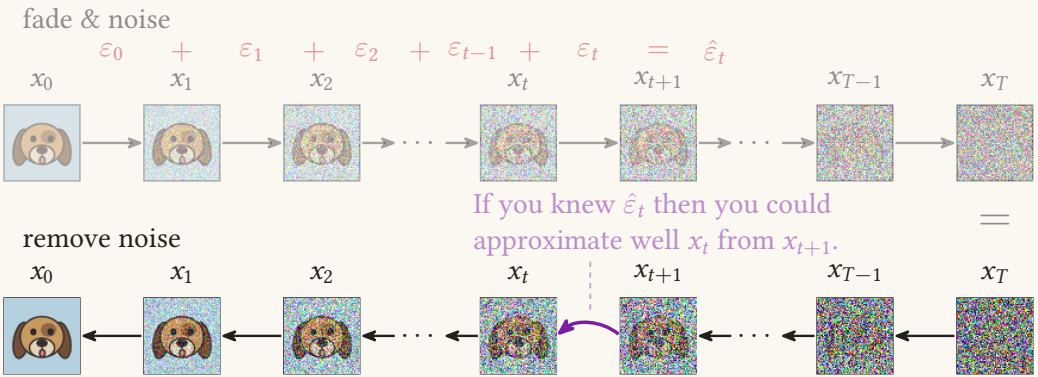


If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

=

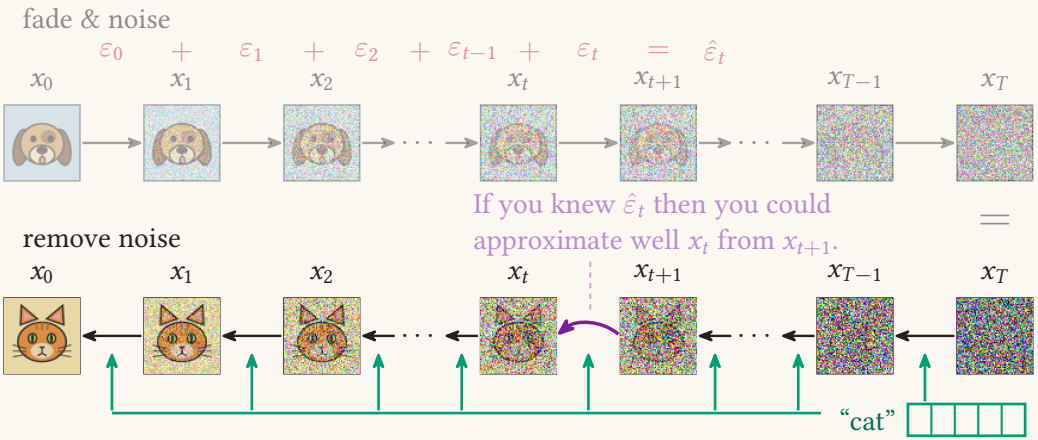
Conditional diffusion models

The prompt "cat" is given to every backward step, so the image it produces is a cat.



Conditional diffusion models

The prompt "cat" is given to every backward step, so the image it produces is a cat.



Training a conditional model

Each training image comes with its caption, which is fed to the model as well.

1. Take an image of the dataset, with its caption.

x_0



"dog"

Training a conditional model

Each training image comes with its caption, which is fed to the model as well.

1. Take an image of the dataset, with its caption.
2. Pick $t \leq T$ at random.

x_0

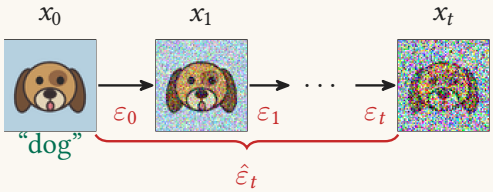


"dog"

Training a conditional model

Each training image comes with its caption, which is fed to the model as well.

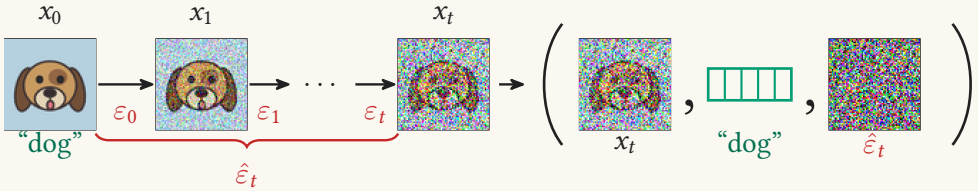
1. Take an image of the dataset, with its caption.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Training a conditional model

Each training image comes with its caption, which is fed to the model as well.

1. Take an image of the dataset, with its caption.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Repeat for many captioned images.